Supplementary Material for the paper "SimFusion: Multi-Fidelity Simulation-based Testing of Automated Driving Systems"

This supplementary material provides visualizations and extended results for the SimFusion framework described in the main paper. The experiments have been conducted on a 16 Core CPU PC with NVIDIA GeForce Ti 4090 and 32 GB RAM.

A. Case Studies

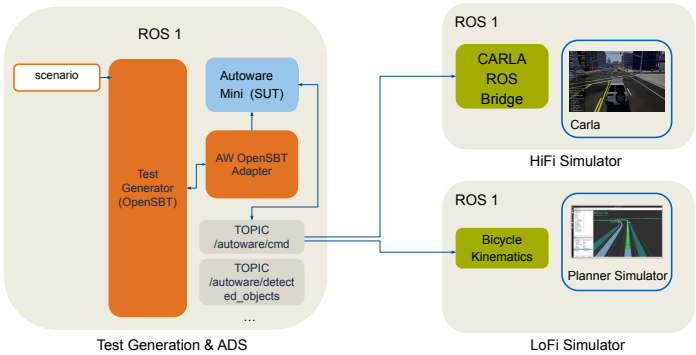


Fig. 1: Autoware case study deployment setup. All systems, i.e. SimFusion Autoware Mini with Carla, and Autoware Mini with LoFi Simulator are deployed in designated docker containers. Test execution is enabled by sending command via ROS topics to the SUT and consuming telemetry data available in the ROS network.

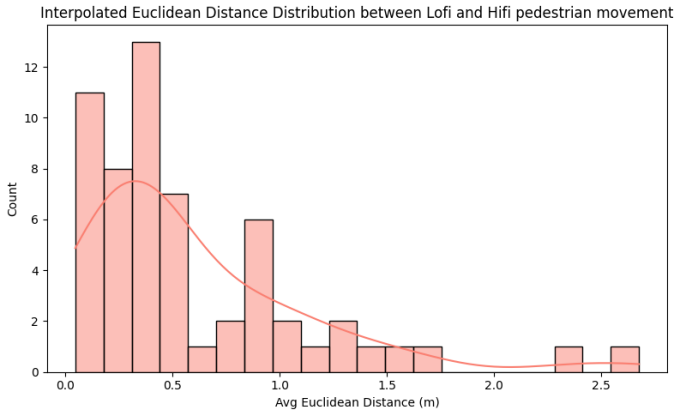


Fig. 2: (Autoware) Scenario alignment validation by measure the average Eucldean distance between a pedestrian simulated in LoFi and HiFi for one and the same scenario sampled over 100 scenarios.

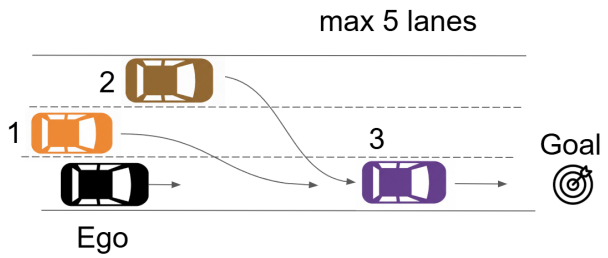
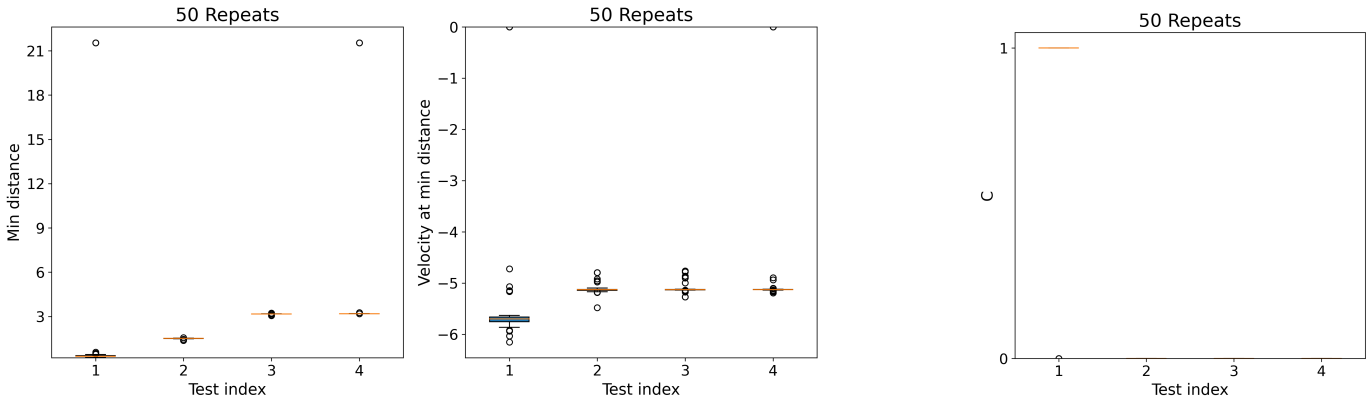


Fig. 3: Overview of scenario modelled in the FrenetiX case study. The scenario is defined by 11 parameters, among those the number of lanes and the position of surrounding vehicles and the goal is controlled.

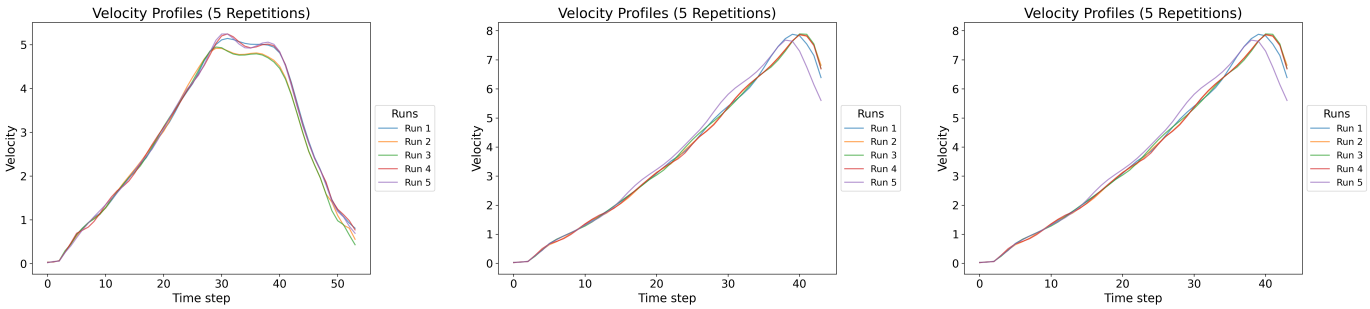
B. Flakiness Analysis (Autoware)



(a) Fitness Functions

(b) Criticality

Fig. 4: (Autoware) Fitness variation by reexecuting randomly generated scenarios 50 times in the Carla-based HiFi simulator. Both distance fitness and velocity fitness variation are low and now variation of the tests outcome (Figure b)) can be observed indicating low simulation flakiness.



(a) Velocity profile 1

(b) Velocity profile 2

(c) Velocity profile 2

Fig. 5: (Autoware) Overview of velocity profiles from reexecution of random scenario 5 times. The profiles reveal no significant deviation as indicated by the low fitness score variation in Figure 4.

C. Effectiveness Test Results

TABLE I: Statistical comparison of SimFusion against baselines using failure rate (ratio) with Mann–Whitney U test ($\alpha = 0.05$) and Vargha–Delaney effect size $\hat{A}_{12}$.

Case Study	**Comparison**	**U**	**p-value**	$\hat{A}_{12}$
Autoware	SimFusion vs LoFi	36	< 0.001	1.00
	SimFusion vs HiFi	36	< 0.001	1.00
	SimFusion vs Surrogate	28	≈ 0.015	0.78
Frenetix	SimFusion vs LoFi	36	< 0.001	1.00
	SimFusion vs HiFi	30	≈ 0.01	0.83
	SimFusion vs Surrogate	30	≈ 0.01	0.83